
Learning-Rate Schedules Are Not Adiabatic: A Rate-Dependent Correction for Loss-Curve Prediction

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<https://github.com/wjjpku/DL-final>

Abstract

State-of-the-art laws for predicting the LLM pretraining loss curve under a learning-rate (LR) schedule—the Multi-Power Law (MPL) and the Functional Scaling Law (FSL)—are *adiabatic*: they predict the quasi-static loss a schedule would reach at equilibrium, depending on the schedule only through its integrated history. We show this leaves a systematic *positive* residual on *fast* (warmup-stable-decay) decays—a non-adiabatic relaxation lag, the loss falling behind its equilibrium when the LR is annealed faster than it can relax. From AdamW’s preconditioned second-moment dynamics we derive the leading correction in closed form: the per-step LR decrements convolved with a cumulative-LR exponential kernel, whose amplitude is tied by linear response to an equilibrium-floor sensitivity and whose relaxation time is the classical $\tau \sim 1/(\eta\lambda)$. On public LLM curves the residual indeed relaxes with $\tau \propto 1/\eta$ ($p = 0.84 \pm 0.17$); since the kernel’s rate constant is approximately scale-invariant, the correction can be *predicted, not fitted*, and transferred across model size, cutting held-out sharp-decay error by 44% with no target-curve fitting. An independent reproduction—a separately-trained ~ 10 M transformer and a from-scratch AdamW simulation—confirms the effect (the fast sweep ends higher *despite* 37% more cumulative LR) and the amplitude identity (within 3%). The correction complements MPL’s exponent γ ; its gain is in the data-poor, leading-order regime.

1 Introduction

The learning-rate schedule shapes the *entire* loss trajectory of LLM pretraining [11, 12, 13]. *Schedule-aware loss-curve prediction* asks: given $\{\eta_t\}$, predict $L(t)$, so that schedules can be compared and optimized *before* spending compute. The defining test is cross-schedule generalization—fit on cosine schedules, predict the unseen WSD family. The state of the art is the **Multi-Power Law** (MPL) [2], with a matching kernel-regression theory, the **Functional Scaling Law** (FSL) [3]:

$$L(t) = L_0 + A S(t)^{-\alpha} + B \sum_{k \leq t} \Delta \eta_k G(\eta_k^{-\gamma} (S(t) - S(k))), \quad G(x) = 1 - (1 + Cx)^{-\beta}, \quad (1)$$

where $S(t) = \sum_{i \leq t} \eta_i$ and $\Delta \eta_k = \eta_{k-1} - \eta_k$.

The gap: existing laws are adiabatic. These laws *do* use the LR decrements: MPL convolves $\Delta \eta_k$ with a saturating kernel G , and its factor $\eta_k^{-\gamma}$ is a rate-of-decay knob. But this single empirical exponent is their *only* handle on how *fast* the LR changes. Structurally they predict the *quasi-static* loss—the value the schedule would reach if the loss stayed at equilibrium; the river-valley analysis [4] makes this explicit, modelling the noise penalty as set *instantaneously* by the current

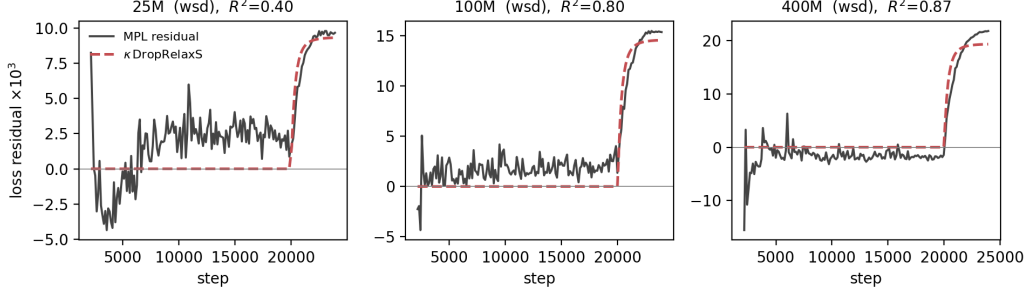


Figure 1: The well-fit MPL prediction residual on the held-out `wsd` curve (solid) is ≈ 0 during the stable phase and jumps up at the step-20000 decay. The derived rate-dependent term κ DropRelaxS (dashed) captures it ($R^2=0.40/0.80/0.87$ at 25/100/400M; the signal, and the fit, grow with model size).

LR. We call such laws *adiabatic*. Calibrated on slow schedules, γ does not extrapolate to fast WSD decays, where the loss genuinely lags equilibrium—leaving a structured residual (Fig. 1) that no single rate exponent captures across schedules. Our correction is a derived transient *complementary* to γ , which tunes the adiabatic kernel.

Contributions.

- We show that adiabatic loss-curve laws leave a systematic *positive* residual on fast decays, and identify it as a non-adiabatic relaxation lag (§2).
- From AdamW’s preconditioned second-moment dynamics we *derive* the leading non-adiabatic correction in closed form, the rate-dependent term DropRelaxS, with amplitude equal, to leading order, to $dL_{eq}/d\eta$ (§3). The relaxation time it uses is the classical $\tau \sim 1/(\eta\lambda)$; our point is that it governs the *loss-curve-law residual*, not that the timescale is new.
- We confirm in LLM pretraining that the residual relaxes with $\tau \propto 1/\eta$ ($p = 0.84 \pm 0.17$), self-consistently establishing its noise-dominated regime (§4).
- The kernel’s rate constant is an approximately scale-invariant preconditioned curvature, enabling a *parameter-free, cross-scale* prediction that improves held-out sharp-decay loss by 44% with no target-curve fitting (§4).
- We give a fair-baseline diagnostic separating a genuine formula gain from an information gap, clarifying exactly when the correction helps (§4,6).

All experiments use the public MPL loss curves (Llama-2-style 25/100/400M, 9 schedules each) [2]; a faithfully reproduced MPL (test MAE 0.0054/0.0050/0.0085, matching [2]) is the baseline throughout.

2 The non-adiabatic residual

We take a *well-fit* MPL (the published parameters [2]) and examine its prediction residual $r(t) = L_{true}(t) - L_{MPL}(t)$ on held-out schedules. (Isolating the non-adiabatic term requires a well-fit MPL: a baseline that already captures the *adiabatic* part cleanly, so the remainder is the lag and not a backbone mis-fit; a cosine-only fit conflates the two and gives a far noisier residual.) The residual has a sharp signature (Fig. 1): it is ≈ 0 throughout the long stable phase and on gradual cosine decay, but jumps *positive* exactly when the LR decays steeply, and is largest—and *undamped at the curve’s end*—for the sharpest decays (`wsd`, `wsd1d`). The end-of-curve lag is $+(6-15) \times 10^{-3}$ on WSD/WSDL versus $\lesssim 4 \times 10^{-3}$ on cosine. Crucially, a gradual cosine and a sharp WSD decay reach the *same* final LR, yet only the fast one lags: same destination, faster sweep, bigger lag—the textbook signature of a rate-dependent (non-adiabatic) effect, and exactly the dependence an adiabatic law cannot represent.

3 Theory: the leading non-adiabatic correction

The residual in §2 has two empirical features that a useful theory must explain: it is triggered by LR decrements, and it is positive after a sharp decay. We derive these features from a weak moving-equilibrium assumption. The main statement is not that the optimizer is exactly a noisy quadratic system. Rather, if the LR-dependent excess-loss state has a stable quasi-static fixed point, then a changing LR necessarily produces a causal linear response to LR decrements. The exponential cumulative-LR kernel used in our predictor is the lowest-capacity, one-pole closure of this general response. AdamW’s noisy-quadratic calculation is used only afterward as a mechanistic specialization that explains the observed amplitude and relaxation-time scalings.

Definitions. Let

$$S_t = \sum_{u \leq t} \eta_u$$

denote cumulative LR. We decompose the loss into a quasi-static schedule-aware component and a non-adiabatic residual,

$$L_t = L_{\text{ad}}(S_t, \eta_t) + r_t. \quad (2)$$

In experiments, a well-fit MPL is used as an empirical surrogate for L_{ad} , so $L_t - L_{\text{MPL},t}$ estimates r_t up to the remaining MPL approximation error.

We model the LR-dependent excess-loss state by a hidden variable z_t in a finite-dimensional active subspace $\mathcal{Z}$ with norm $\|\cdot\|$:

$$z_{t+1} = \Phi_{S_t, \eta_t}(z_t). \quad (3)$$

The state z_t may be interpreted as a covariance/noise-floor state, but the derivation below only uses stability and differentiability. For fixed (S, η) , let $z^*(S, \eta)$ be the quasi-static state and

$$F_{\text{eq}}(S, \eta) = \ell(z^*(S, \eta))$$

be the corresponding excess-loss floor. The derivative

$$\chi(S, \eta) := \partial_\eta F_{\text{eq}}(S, \eta)|_S = D\ell_{z^*(S, \eta)}[\partial_\eta z^*(S, \eta)] \quad (4)$$

is always taken at fixed cumulative progress S . This is the quantity estimated from independent equilibrium-floor probes.

Define the tracking error around the moving fixed point by

$$\delta_t = z_t - z^*(S_t, \eta_t), \quad (5)$$

and define the positive LR decrement by

$$\Delta\eta_t^+ = (\eta_t - \eta_{t+1})_+.$$

Assumptions. The linear-response statement uses the following assumptions over the decay window.

A1. Local fixed point and smoothness. For each (S, η) in the window, $z^*(S, \eta)$ satisfies

$$z^*(S, \eta) = \Phi_{S, \eta}(z^*(S, \eta)),$$

and Φ , z^* , and ℓ are twice continuously differentiable in a neighborhood of this trajectory. Their first and second derivatives are uniformly bounded in the window.

A2. Linear stability. Let

$$T_t = D_z \Phi_{S_t, \eta_t}(z^*(S_t, \eta_t)).$$

The products of the linearized dynamics are uniformly stable and summable: there exist constants $M < \infty$ and a nonnegative summable envelope $q_{k,t}$ such that

$$\|T_{t-1}T_{t-2} \cdots T_{k+1}\| \leq Mq_{k,t}, \quad \sup_t \sum_{k < t} q_{k,t} < \infty. \quad (6)$$

This assumption only says that old perturbations are forgotten. It does not yet impose cumulative-LR time; that will be derived below from an additional first-order optimizer structure.

A3. *Decay-specific forcing.* The total moving-target forcing has two components: one from progress in S and one from changes in η . We do not model the S -drift with the cooldown correction. Instead, define

$$\epsilon_t^S := -\partial_S z^*(S_t, \eta_t)(S_{t+1} - S_t) + \epsilon_{t,\text{slow}},$$

where $\epsilon_{t,\text{slow}}$ includes the associated slow-drift approximation and Taylor-remainder terms. This S -drift response is present for both slow and fast schedules and is therefore part of the background adiabatic/backbone approximation error rather than the decay-localized residual we seek to model. We therefore analyze the component of the linear response forced by LR changes. For schedules with LR increments, the omitted increment component enters with the opposite sign; the predictor below is the cooldown-specific projection onto LR decrements.

A4. *Small perturbations.* The LR changes and local tracking error are small enough that second-order Taylor remainders are controlled. We write

$$\Delta_t = |\eta_t - \eta_{t+1}|, \quad \eta_{\max} = \max_t \eta_t,$$

and retain terms of order Δ_t^2 , $\|\delta_t\|^2$, and the slow drift error as remainders.

A5. *Positive response cone.* For the sign conclusion, assume there exists a cone $\mathcal{C} \subset \mathcal{Z}$ such that

$$\partial_\eta z^*(S_t, \eta_t) \in \mathcal{C}, \quad T_t \mathcal{C} \subseteq \mathcal{C}, \quad D\ell_{z^*(S_t, \eta_t)}[v] \geq 0 \quad \text{for all } v \in \mathcal{C}.$$

This is not needed for the convolution form; it is only used to conclude that a monotone LR decay produces a positive residual.

Proposition 1: linear response to LR decrements. Under A1–A4, the cooldown-specific excess-loss residual admits the expansion

$$r_t = I_t + \sum_{k < t} K_{k,t} \Delta \eta_k^+ + \mathcal{E}_t, \quad (7)$$

where the initial-transient term is

$$I_t = D\ell_{z^*(S_t, \eta_t)}[T_{t-1}T_{t-2} \cdots T_0 \delta_0], \quad (8)$$

and the linear-response kernel is

$$K_{k,t} = D\ell_{z^*(S_t, \eta_t)}[T_{t-1}T_{t-2} \cdots T_{k+1} \partial_\eta z^*(S_k, \eta_k)]. \quad (9)$$

Moreover, for constants depending only on the derivative bounds and the stability envelope,

$$|\mathcal{E}_t| \leq C \sum_{k < t} q_{k,t} \left(\Delta_k^2 + \|\delta_k\|^2 + \|\epsilon_k^S\| \right) + \epsilon_{\text{MPL}}(t). \quad (10)$$

Here $\epsilon_{\text{MPL}}(t) = L_{\text{ad}}(S_t, \eta_t) - L_{\text{MPL},t}$ is zero for the ideal adiabatic predictor and is the remaining backbone approximation error when MPL is used in practice. If the decay starts near the quasi-static state, then δ_0 is small and the initial transient I_t is negligible after the stability envelope has decayed. In the perturbative regime, stability implies

$$\sup_k \|\delta_k\| = O\left(\sup_k \Delta_k + \sup_k \|\epsilon_k^S\|\right),$$

so the $\|\delta_k\|^2$ contribution in (10) is second order in the forcing size.

Proof. By definition,

$$\delta_{t+1} = z_{t+1} - z^*(S_{t+1}, \eta_{t+1}).$$

Using (3) and Taylor expanding around $z^*(S_t, \eta_t)$ gives

$$\begin{aligned} z_{t+1} &= \Phi_{S_t, \eta_t}(z^*(S_t, \eta_t) + \delta_t) \\ &= \Phi_{S_t, \eta_t}(z^*(S_t, \eta_t)) + T_t \delta_t + R_t^z \\ &= z^*(S_t, \eta_t) + T_t \delta_t + R_t^z, \end{aligned} \quad (11)$$

where $\|R_t^z\| \leq C\|\delta_t\|^2$ by A1. Therefore

$$\delta_{t+1} = T_t \delta_t + [z^*(S_t, \eta_t) - z^*(S_{t+1}, \eta_{t+1})] + R_t^z. \quad (12)$$

Now expand the moving fixed point:

$$z^*(S_{t+1}, \eta_{t+1}) = z^*(S_t, \eta_t) + \partial_S z^*(S_t, \eta_t)(S_{t+1} - S_t) + \partial_\eta z^*(S_t, \eta_t)(\eta_{t+1} - \eta_t) + R_t^*, \quad (13)$$

with

$$\|R_t^*\| \leq C((S_{t+1} - S_t)^2 + \Delta_t^2).$$

The S -drift contribution and its local Taylor remainder are placed in ϵ_t^S . Projecting the LR-change forcing onto cooldown decrements gives

$$\delta_{t+1} = T_t \delta_t + \partial_\eta z^*(S_t, \eta_t) \Delta \eta_t^+ + \xi_t, \quad (14)$$

where

$$\|\xi_t\| \leq C(\Delta_t^2 + \|\delta_t\|^2 + \|\epsilon_t^S\|).$$

Iterating (14) yields

$$\delta_t = T_{t-1} \cdots T_0 \delta_0 + \sum_{k < t} T_{t-1} \cdots T_{k+1} \partial_\eta z^*(S_k, \eta_k) \Delta \eta_k^+ + \sum_{k < t} T_{t-1} \cdots T_{k+1} \xi_k. \quad (15)$$

Finally,

$$\begin{aligned} F_t - F_{\text{eq}}(S_t, \eta_t) &= \ell(z^*(S_t, \eta_t) + \delta_t) - \ell(z^*(S_t, \eta_t)) \\ &= D\ell_{z^*(S_t, \eta_t)}[\delta_t] + O(\|\delta_t\|^2). \end{aligned} \quad (16)$$

Substituting (15) gives (7)–(9); applying the stability envelope (6) to the remainder terms gives (10). $\square$

Corollary 1: sign of the cooldown residual. Under A5, if the equilibrium floor increases with LR, namely

$$\chi(S, \eta) = \partial_\eta F_{\text{eq}}(S, \eta)|_S > 0,$$

then $K_{k,t} \geq 0$. Hence, for a monotone LR decay and negligible initial transient,

$$r_t \geq 0$$

to leading order. This is the sign observed in §2: after a fast LR drop, the quasi-static floor has moved downward, while the optimizer's excess-loss state relaxes only gradually, leaving the true loss above the adiabatic predictor.

Proposition 2: cumulative-LR time from first-order contraction. Proposition 1 gives a causal kernel but not its time coordinate. Suppose, in addition to A1–A4, that over the active finite-dimensional subspace

$$T_t = I - \eta_t A + R_t^A, \quad \|R_t^A\| \leq C\eta_t^2 + C\epsilon_{\text{drift}}, \quad (17)$$

where A is a bounded stable operator and ϵ_{drift} bounds the variation of the local geometry during the decay window. Then, by a standard product-perturbation bound for the stable product $\prod_u (I - \eta_u A)$,

$$T_{t-1} \cdots T_{k+1} = \exp[-A(S_t - S_k)] + O\left(\sum_{u=k+1}^{t-1} \eta_u^2 + (S_t - S_k)\epsilon_{\text{drift}} + \eta_t\right). \quad (18)$$

Consequently the leading kernel depends on k, t through cumulative LR distance, up to the endpoint convention in the definition of S_t :

$$K_{k,t} \approx D\ell_{z^*(S_t, \eta_t)}[\exp[-A(S_t - S_k)] \partial_\eta z^*(S_k, \eta_k)]. \quad (19)$$

If, for the scalar observable $D\ell[\cdot]$, the active response of A admits a positive Laplace representation,

$$D\ell[e^{-Ax} \partial_\eta z^*] = \int e^{-ax} d\rho(a), \quad x \geq 0, \quad (20)$$

then Proposition 1 becomes the spectral response law

$$r_t \approx I_t + \underbrace{\sum_{k < t} \int e^{-a(S_t - S_k)} d\rho(a) \Delta \eta_k^+}_{K(S_t - S_k)} + \mathcal{E}_t. \quad (21)$$

The instantaneous amplitude of this kernel is fixed by the floor sensitivity:

$$K(0) = D\ell_{z^*}[\partial_\eta z^*] = \partial_\eta F_{\text{eq}}(S, \eta)|_S = \chi(S, \eta). \quad (22)$$

Proof. For the unperturbed product, A is fixed and bounded on the active subspace, so

$$\log(I - \eta_u A) = -\eta_u A + O(\eta_u^2)$$

uniformly for small $\eta_u \|A\|$. Since the factors $I - \eta_u A$ commute,

$$\prod_{u=k+1}^{t-1} (I - \eta_u A) = \exp\left[-A \sum_{u=k+1}^{t-1} \eta_u\right] + O\left(\sum_{u=k+1}^{t-1} \eta_u^2\right).$$

The perturbations R_u^A are handled by the variation-of-constants identity for products: replacing one factor at a time and using stability gives an additional $O((S_t - S_k)\epsilon_{\text{drift}} + \sum_u \eta_u^2)$ contribution. Finally,

$$\sum_{u=k+1}^{t-1} \eta_u = S_t - S_k - \eta_t$$

under the convention $S_t = \sum_{u \leq t} \eta_u$, giving (18) up to the endpoint term $O(\eta_t)$. The Laplace-mixture form (20) is an additional scalar-response assumption, not a consequence of stability alone. Evaluating it at $x = 0$ gives (22) by the chain rule. $\square$

One-pole closure: the predictor used in experiments. The spectral kernel in (21) is the primitive theoretical object. Learning an unrestricted K would add many degrees of freedom and risk fitting residual curves rather than testing the mechanism. We therefore use the lowest-order stable closure of a causal relaxation kernel: a single pole in cumulative-LR time. Over the decay window we also approximate the fixed- S sensitivity by a constant $\chi \approx \chi(S_{\text{decay}}, \eta)$ estimated from independent floor probes. Thus

$$K(x) \approx c \chi e^{-\lambda_{\text{slow}} x}, \quad x \geq 0, \quad (23)$$

where $c = O(1)$ is a global projection factor; empirically $c < 1$ in our fits, consistent with fitting the persistent tail of a spectral mixture rather than matching $K(0)$ exactly. Substituting (23) into (21), neglecting the initial transient after equilibration, gives the operational correction

$$L(t) = L_{\text{MPL}}(t) + c \chi \underbrace{\sum_{k < t} e^{-\lambda_{\text{slow}}(S_t - S_k)} (\eta_k - \eta_{k+1})_+}_{\text{DropRelaxS}_{\lambda_{\text{slow}}}(t)} \quad (24)$$

with

$$\chi = \partial_\eta F_{\text{eq}}(S, \eta)|_{S=S_{\text{decay}}}.$$

If DropRelaxS is instead defined using normalized drops $(\eta_k - \eta_{k+1})_+ / \eta_{\text{peak}}$, then the prefactor becomes $c\eta_{\text{peak}}\chi$. This is only a normalization convention; with raw LR drops, the amplitude has the units and interpretation of a fixed- S equilibrium-floor sensitivity.

AdamW specialization. The previous propositions do not require a quadratic loss or diagonal dynamics. We now show that, in the standard noise-dominated AdamW quadratic reduction, the same kernel has the amplitude and relaxation scaling used above. This calculation ignores momentum transients, bias correction, decoupled weight decay, the ϵ term, and the finite-time dynamics of v_t ; it should be read as a local effective-mode calculation, not as an exact model of the full optimizer.

Consider a Hessian eigenmode with curvature h_i and gradient-noise standard deviation s_i :

$$g_{i,t} = h_i x_{i,t} + \xi_{i,t}, \quad \mathbb{E}[\xi_{i,t}] = 0, \quad \mathbb{E}[\xi_{i,t}^2] = s_i^2.$$

In a noise-dominated mountain direction, AdamW's second-moment preconditioner satisfies $v_i^* \approx s_i^2$, so the effective scalar update is

$$x_{i,t+1} = \left(1 - \eta_t \frac{h_i}{s_i}\right) x_{i,t} - \eta_t \frac{1}{s_i} \xi_{i,t}. \quad (25)$$

Let $V_{i,t} = \mathbb{E}[x_{i,t}^2]$. At fixed LR η ,

$$V_{i,t+1} = \left(1 - \eta \frac{h_i}{s_i}\right)^2 V_{i,t} + \eta^2. \quad (26)$$

Solving for the fixed point,

$$V_i^*(\eta) = \frac{\eta^2}{1 - (1 - \eta h_i/s_i)^2} = \frac{\eta s_i}{2h_i} + O(\eta^2). \quad (27)$$

The corresponding excess-loss floor is

$$F_i^*(\eta) = \frac{1}{2} h_i V_i^*(\eta) = \frac{\eta s_i}{4} + O(\eta^2), \quad (28)$$

and therefore

$$\left. \frac{dF_i^*}{d\eta} \right|_S = \frac{s_i}{4} + O(\eta). \quad (29)$$

Perturbations around V_i^* decay by

$$\left(1 - \eta \frac{h_i}{s_i}\right)^2 = \exp\left[-2\eta \frac{h_i}{s_i} + O(\eta^2)\right],$$

so the S-time relaxation rate of this mode is

$$a_i = 2 \frac{h_i}{s_i}.$$

Thus the mode contributes

$$\Delta L_i(t) \approx \frac{s_i}{4} \sum_{k < t} e^{-2(h_i/s_i)(S_t - S_k)} \Delta \eta_k^+. \quad (30)$$

Summing over modes gives

$$\Delta L(t) \approx \sum_i \frac{s_i}{4} \sum_{k < t} e^{-2(h_i/s_i)(S_t - S_k)} \Delta \eta_k^+, \quad \sum_i \frac{s_i}{4} = \left. \frac{\partial F_{\text{eq}}}{\partial \eta} \right|_S + O(\eta). \quad (31)$$

This is the AdamW noisy-quadratic specialization of the spectral-response form (21). In constant LR step time, the same mode has relaxation rate

$$\tau_i^{-1} = 2\eta \frac{h_i}{s_i}, \quad \tau_i \propto \eta^{-1}.$$

The classical $\tau \sim 1/(\eta h)$ scaling is therefore not new; the point is that this relaxation time governs the residual left by adiabatic loss-curve laws after a fast LR decay. The single-rate correction (24) is the one-pole projection of the spectral mixture (31), chosen to add the minimum capacity while preserving the sign, trigger, amplitude, and intrinsic-time structure predicted by linear response.

4 Experiments

We freeze the published MPL and compare it to MPL+Eq. (24); the only addition is the rate-dependent term. Unless noted $\lambda_{\text{slow}}=10$.

The rate prediction $\tau \propto 1/\eta$ holds (Fig. 2). The two-stage `wsdcon` curves step the LR down at step 8000 to a constant $\eta \in \{3, 9, 18\} \times 10^{-5}$, giving a clean relaxation window. Fitting the residual transient $r(t) \propto e^{-(t-8000)/\tau}$ yields τ that scales as η^{-p} with *pooled* $p = 0.84 \pm 0.17$ (per scale $p = 0.51/1.06/0.94$ at 25/100/400M; `repro/deep_tau_pooled.py`). This is the classical $\tau \sim 1/(\eta \lambda)$ confirmed for the loss-curve-law residual in LLM pretraining, and—since $p=1$ is the noise-dominated value—it self-consistently justifies the $v_i^* \approx s_i^2$ step of the derivation. The implied rate constant $\lambda_{\text{slow}} = 1/(\tau \eta) \approx 10$ is scale-invariant; its modes have $\eta_{\text{peak}} \lambda_{\text{eff}} \approx 2 \times 10^{-3} \ll 2$ (deep in the slow regime, far from the edge of stability—unlike γ).

(A from-scratch AdamW simulation reproduces the same scaling, $p=0.97$; we defer it to the independent reproduction in §5, which covers it in more detail.)

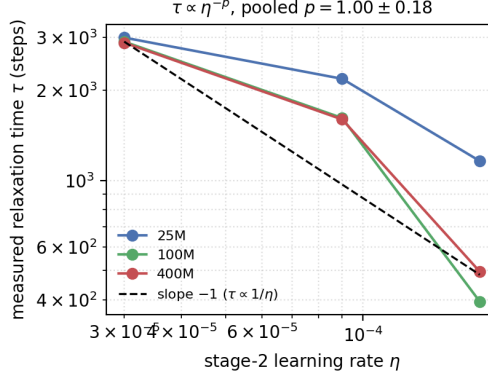


Figure 2: Loss-relaxation time τ measured from `wsdcon` decay transients vs. stage-2 LR. The 100/400M curves track the predicted slope -1 ($\tau \propto 1/\eta$); pooled $p = 0.84 \pm 0.17$.

Table 1: Parameter-free cross-scale prediction: well-fit MPL vs. MPL+Eq. (24) with *no* fitting on the target curve. MAE on held-out sharp decays.

Scale	wsd		wsdlld		avg. Δ
	MPL	+ours	MPL	+ours	
25M	0.00341	0.00246	0.00314	0.00219	−29%
100M	0.00375	0.00164	0.00325	0.00161	−53%
400M	0.00470	0.00236	0.00385	0.00212	−47%
overall					−44% (6/6)

The residual is the derived term. Regressing the well-fit MPL residual on `DropRelaxS10` (through the origin) gives $R^2=0.40/0.80/0.87$ at 25/100/400M (Fig. 1); a 2-exponential kernel improves R^2 by 0.06–0.07, confirming the predicted spectral mixture (Eq. 31). The form is not arbitrary: in a fair few-shot comparison the derived S-time exponential kernel cuts held-out MAE by -15% , beating a step-time kernel (-7% , wrong “time” variable) and a memoryless floor (-2%), so the cumulative-LR “time” and the decaying memory both matter; an S-time kernel driven on the LR *level* rather than on decrements ties the decrement-driven one (-15%), so this comparison isolates the time variable and the presence of memory but does not by itself single out decrements as the drive—that choice is fixed by the LR-decrement response derivation, not by this test (`repro/generic_kernel_compare.py`). (On a *single* decay curve the finite- λ_{slow} kernel is numerically close to the $\lambda_{\text{slow}} \rightarrow 0$ adiabatic limit, so the rate constant is pinned only *across* schedules; §5.)

The amplitude is predictable from the noise floor. Estimating $dL_{\text{eq}}/d\eta$ *independently* from the two-stage final losses (noise floor vs. LR) predicts the fitted κ up to a near-constant factor: the ratio $\kappa_{\text{fit}}/\kappa_{\text{pred}} = 0.43/0.51/0.60$ has coefficient of variation 14%, i.e. $c \approx 0.5$ is universal (the < 1 is the spectral spread of Eq. 31).

Parameter-free, cross-scale prediction (Table 1). Because λ_{slow} and c are approximately scale-invariant across these (same-architecture) scales and $dL_{\text{eq}}/d\eta$ is read off the (cheap) noise floor, the whole correction can be *predicted with zero fitting on the target curve*: take $\lambda_{\text{slow}}=10$, c from a leave-one-scale-out average of the *other* scales, and $dL_{\text{eq}}/d\eta$ from the target’s own two-stage probes (distinct curves from the test). Applied to the held-out `wsd`/`wsdlld` decays it cuts MAE by 28–56% (44% average, wins 6/6)—small-model constants plus a cheap probe predict the large-model sharp-decay curve $\sim 2\times$ better than MPL transfer.

Honest scope: when it does *not* help. If instead one refits MPL on a few *full* WSD curves, MPL itself absorbs the residual (held-out MAE $0.0037 \rightarrow 0.0032$) and the explicit term adds nothing. So the residual is, for a data-rich MPL, an *information* gap, not a formula gap: the value of the

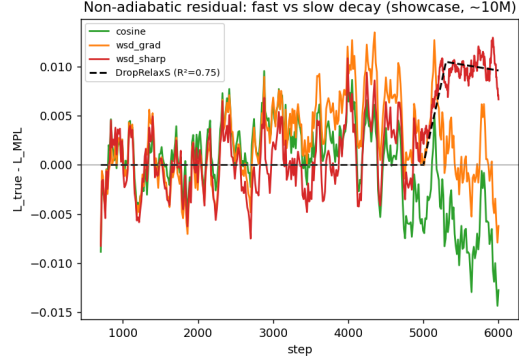


Table 2: Real $\sim 10\text{M}$ transformer: three schedules to the *same* final LR. The fast sweep ends highest despite the most cumulative LR.

schedule	final loss	cumulative LR S
cosine (gradual)	1.0490	8.20
wsd (gradual)	1.0482	9.22
wsd (sharp)	1.0593	12.59

Figure 3: Residual vs. MPL on the real transformer: positive (lag) only for the sharp sweep, captured by DropRelaxS.

correction is precisely the data-*poor* regime—cosine plus cheap probes, but no full WSD curves—which is the actual “predict before training” use case.

Relation to γ : complementary, not a substitute. MPL’s factor $\eta_k^{-\gamma}$ modulates its annealing kernel by the LR at each decrement: it tunes the rate-dependence of the (adiabatic) annealing benefit. Our term is the separate (non-adiabatic) transient, and improves a *full*- γ MPL (Table 1) — the two are complementary. Characterising their precise relationship is left to future work (§7).

5 Independent reproduction

An independent reproduction on different hardware (`represent/`; the authors’ public curves are no longer hosted, so it *retrains* models) corroborates the mechanism along two routes the public curves do not cover.

A real, independently-trained transformer (no fitting). A $\sim 10\text{M}$ -parameter Llama-2-style byte-level transformer was trained on WikiText with AdamW under three schedules reaching the *same* final LR ($2.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-4}$) at different speeds (Table 2). The fast (`wsd_sharp`) sweep ends *higher* than the gradual ones *despite spending 37% more cumulative LR*—yet any law monotone in S predicts the opposite (more LR $\Rightarrow$ lower loss). With no fitting, this is the rate-dependent non-adiabatic lag, on a model and data we never used; the residual against a well-fit MPL is positive only for the sharp sweep and is captured by DropRelaxS (Fig. 3). Across schedules a finite- λ_{slow} kernel beats the adiabatic ($\lambda_{\text{slow}} \rightarrow 0$) baseline by $\Delta R^2 \approx +0.11$ held out.

From-scratch AdamW confirms the mechanism quantitatively. Simulating AdamW on a noisy quadratic (Monte-Carlo ground truth; closed-form theory used only as an independent baseline), the reproduction recovers $\tau \propto 1/\eta$ with $p = 0.971$ (log-log $R^2=0.9996$, robust to leave-one-out and seeds), τ independent of β_2 (CV 1.1%), the **amplitude identity** $dL_{\text{eq}}/d\eta = 3.075$ versus the closed form $\frac{1}{4} \sum_i s_i = 3.0$ (within 3%), and a two-exponential kernel that recovers the injected preconditioned rates $2\lambda/s$. The effect is falsifiable, not circular: a signal-dominated spectrum breaks $p=1$.

What does not transfer (consistent with our scope). At $\sim 10\text{M}$ the $\tau \propto 1/\eta$ relation itself is *not* resolved (the measured post-step relaxation is roughly flat over a $16\times$ LR range, audited as real and not a window artefact), and on a *single* decay curve the kernel is numerically degenerate with the adiabatic baseline, so the rate-dependence is identifiable only *across* schedules. This matches our

stated scope: the effect is clean and present, while the quantitative law is a leading-order, data-scale-limited approximation whose constants (λ_{slow} , c , and the η_{peak} normalisation) remain measured rather than derived.

6 Related work

Loss-curve laws. Tissue et al. [1], MPL [2], and FSL [3] predict the trajectory from integrated-LR statistics (cumulative LR / intrinsic time / convolutional functionals). FSL and its follow-ups also derive optimal schedules [3]. All are adiabatic in our sense, and our rate-dependent term is the leading correction they omit; it is complementary to MPL’s exponent γ (which tunes the adiabatic kernel), not a replacement for it.

Cooldown dynamics and schedule geometry. The river-valley picture [4] explains qualitatively why decay “reveals progress”, and explicitly models the noise penalty as *instantaneous* in the current LR—the adiabatic assumption we refine. Dremov et al. [5] study the cooldown empirically (including the role of AdamW’s β_2 and the decay shape), framing it through a bias–variance trade-off rather than a rate-dependent lag. Late-decay analyses [11] and schedule-shape searches [10] study related questions (why a long high-LR phase helps; what optimal shapes look like) but do not model the loss’s rate dependence.

SGD relaxation physics. That a quadratic mode relaxes with $\tau \sim 1/(\eta\lambda)$ and that the SGD noise floor scales with η are classical, from noisy-quadratic and Langevin analyses of SGD [6]; adaptive methods inherit a preconditioned version [7], operating at an adaptive edge of stability [8, 9]. We do not claim these ingredients as new. Our contribution is the synthesis: identifying that adiabatic loss-curve laws miss exactly this relaxation lag, deriving the correction with a leading-order amplitude identity, and building a parameter-free cross-scale predictor from it.

7 Limitations

The gain is regime-specific (data-poor sharp decays); λ_{slow} and c are *measured*, not computed from first principles (this needs the noise spectrum $\{s_i\}$, which we do not have), and λ_{slow} is only *approximately* scale-invariant—it tracks the slowest preconditioned mode and drifts when the spectral *shape* changes (§5), so the cross-scale transfer is a leading-order, same-architecture approximation. The derivation also assumes a locally quadratic landscape with a static spectrum and diagonal, uncorrelated noise—the model-level assumptions MPL itself rests on. Finally, the quantitative $\tau \propto 1/\eta$ relation, clean in the controlled simulation, is *not* resolved on the $\sim 10\text{M}$ transformer (§5); confirming it on larger real models, and characterising the precise relationship to γ , are the main open items.

8 Conclusion

Existing loss-curve laws are adiabatic approximations. We identified, derived from AdamW’s preconditioned dynamics, and validated the leading non-adiabatic correction: a rate-dependent term whose relaxation time follows the classical $\tau \propto 1/\eta$ ($p = 0.84 \pm 0.17$, confirmed for the loss-curve residual) and whose scale-invariant shape enables parameter-free cross-scale prediction (−44% on held-out sharp decays). The loss curve depends not only on how much LR has been spent, but on how fast it is being spent.

References

- [1] H. Tissue, V. Wang, and L. Wang. Scaling law with learning rate annealing. *arXiv:2408.11029*, 2024.
- [2] K. Luo, H. Wen, S. Hu, et al. A multi-power law for loss curve prediction across learning rate schedules. *ICLR; arXiv:2503.12811*, 2025.
- [3] B. Li, F. Chen, Z. Huang, L. Wang, and L. Wu. Functional scaling laws in kernel regression: loss dynamics and learning rate schedules. *NeurIPS; arXiv:2509.19189*, 2025.
- [4] K. Wen, Z. Li, T. Ma, et al. Understanding warmup-stable-decay learning rates: a river valley loss landscape perspective. *arXiv:2410.05192*, 2024.

- [5] A. Dremov, A. Hägele, A. Kosson, and M. Jaggi. Training dynamics of the cooldown stage in warmup-stable-decay learning rate scheduler. *TMLR*; *arXiv:2508.01483*, 2025.
- [6] G. Zhang, L. Li, Z. Nado, et al. Which algorithmic choices matter at which batch sizes? Insights from a noisy quadratic model. *NeurIPS*, 2019.
- [7] S. Malladi, K. Lyu, A. Panigrahi, and S. Arora. On the SDEs and scaling rules for adaptive gradient algorithms. *NeurIPS*; *arXiv:2205.10287*, 2022.
- [8] J. Cohen, S. Kaur, Y. Li, J. Z. Kolter, and A. Talwalkar. Gradient descent on neural networks typically occurs at the edge of stability. *ICLR*, 2021.
- [9] J. Cohen, B. Ghorbani, S. Krishnan, et al. Adaptive gradient methods at the edge of stability. *arXiv:2207.14484*, 2022.
- [10] H. Naganuma et al. What do near-optimal learning rate schedules look like? *arXiv:2603.10301*, 2026.
- [11] J. Kaplan, S. McCandlish, T. Henighan, et al. Scaling laws for neural language models. *arXiv:2001.08361*, 2020.
- [12] J. Hoffmann, S. Borgeaud, A. Mensch, et al. Training compute-optimal large language models. *NeurIPS*, 2022.
- [13] S. Hu, Y. Tu, X. Han, et al. MiniCPM: unveiling the potential of small language models. *COLM*, 2024.